

Abstract algebra – 5

Representations and the averaging operator

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Theoretical minimum. Olympiads almost never ask for character tables, but they constantly ask for one device: *average over the group*. Everything in this sheet — the projection onto invariants, Burnside’s lemma, unitarisability, Maschke’s theorem, orthogonality of characters — is that single device applied to different objects.

Definitions. A *representation* of a finite group G is a homomorphism $\rho: G \rightarrow \mathrm{GL}(V)$ with V a finite-dimensional complex vector space; $\dim V$ is its *dimension*. A subspace W is *invariant* if $\rho(g)W \subseteq W$ for all g , and ρ is *irreducible* if the only invariant subspaces are 0 and V . The *character* is $\chi_\rho(g) = \mathrm{tr} \rho(g)$; it is constant on conjugacy classes and $\chi(e) = \dim V$. We write $V^G = \{v : \rho(g)v = v \text{ for all } g\}$ for the subspace of invariants, and $\langle \chi, \psi \rangle = \frac{1}{|G|} \sum_g \chi(g) \overline{\psi(g)}$.

Theorem 1 (the averaging operator). Put $P = \frac{1}{|G|} \sum_{g \in G} \rho(g)$. Then $\rho(h)P = P\rho(h) = P$ for every h , because $g \mapsto hg$ only permutes the summands; consequently $P^2 = P$ and the image of P is exactly V^G , while P acts as the identity on V^G . A projection has trace equal to the dimension of its image, so

$$\dim V^G = \mathrm{tr} P = \frac{1}{|G|} \sum_{g \in G} \chi(g).$$

This formula is the whole sheet. In particular, if $V^G = 0$ then $\sum_g \rho(g) = 0$ as a matrix.

Corollary (Burnside’s lemma again). Let G act on a finite set X and let $V = \mathbb{C}^X$ with $\rho(g)$ permuting the coordinates. Then $\chi(g) = |\mathrm{Fix}(g)|$ and V^G has as a basis the indicators of the orbits, so the number of orbits is $\frac{1}{|G|} \sum_g |\mathrm{Fix}(g)|$ — the counting lemma of Abstract algebra – 4, now read as a statement about a trace.

Theorem 2 (unitarisability). Start from any Hermitian inner product $\langle \cdot, \cdot \rangle_0$ on V and average it: $\langle u, v \rangle = \frac{1}{|G|} \sum_g \langle \rho(g)u, \rho(g)v \rangle_0$. This is again a Hermitian inner product and it is G -invariant. Hence every finite subgroup of $\mathrm{GL}_n(\mathbb{C})$ is conjugate to a subgroup of the unitary group; every element of a finite matrix group is diagonalisable and all its eigenvalues are roots of unity.

Theorem 3 (Maschke). If $W \subseteq V$ is invariant, take any projection π onto W and average it: $\tilde{\pi} = \frac{1}{|G|} \sum_g \rho(g)\pi\rho(g)^{-1}$. Then $\tilde{\pi}$ is again a projection onto W and commutes with the action, so $\ker \tilde{\pi}$ is an invariant complement. By induction every complex representation of a finite group is a direct sum of irreducibles. The averaging breaks when $|G|$ is divisible by the characteristic of the field, and the theorem is then false.

Theorem 4 (Schur's lemma). If $T: V \rightarrow W$ intertwines two irreducible representations, then $\ker T$ and $\mathrm{im} T$ are invariant, so $T = 0$ or T is an isomorphism; and if $V = W$ over $\mathbb{C}$, then T has an eigenvalue λ and $T - \lambda I$ is a non-invertible intertwiner, hence $T = \lambda I$.

Fact 1 (orthogonality, in one line). Apply Theorem 1 to the space $\mathrm{Hom}(V, W)$ with the action $g \cdot T = \rho_W(g)T\rho_V(g)^{-1}$, whose character is $\overline{\chi_V}\chi_W$ and whose invariants are the intertwiners. Theorem 1 gives $\dim \mathrm{Hom}_G(V, W) = \langle \chi_W, \chi_V \rangle$, which by Schur's lemma equals 1 for equivalent irreducibles and 0 otherwise: irreducible characters are orthonormal. The regular representation has character $|G|$ at e and 0 elsewhere, whence $\sum_i (\dim V_i)^2 = |G|$, and the number of irreducible representations equals the number of conjugacy classes.

1. Prove that $P = \frac{1}{|G|} \sum_g \rho(g)$ is a projection onto V^G and that $\dim V^G = \frac{1}{|G|} \sum_g \chi(g)$. Deduce Burnside's lemma, and then prove that for the permutation representation $\langle \chi, \chi \rangle$ equals the number of orbits of G on $X \times X$; in particular a transitive action is 2-transitive if and only if $\chi - 1$ is irreducible.
2. Average points instead of operators: prove that a finite group of affine

transformations of $\mathbb{R}^n$ has a common fixed point, and deduce that a finite group of isometries of $\mathbb{R}^n$ is conjugate to a subgroup of the orthogonal group. Show that both statements fail for infinite groups. (The Putnam problem “a finite matrix group with $\sum \operatorname{tr} M_i = 0$ satisfies $\sum M_i = 0$ ”, which is Theorem 1 read backwards, stands in Linear algebra – 4, homework 5.)

3. Prove that every finite subgroup of $\operatorname{GL}_n(\mathbb{C})$ is conjugate to a group of unitary matrices, and deduce that each of its elements is diagonalisable with eigenvalues roots of unity. Then let $G \leq \operatorname{GL}_n(\mathbb{R})$ be a finite group in which $g^2 = I$ for every element: prove that G is abelian and $|G| \leq 2^n$, and describe the groups attaining the bound.
4. Prove Maschke’s theorem by averaging a projection, and show that it fails over $\mathbb{F}_p$: for the group $\mathbb{Z}/p$ acting on $\mathbb{F}_p^2$ by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ the invariant line has no invariant complement.
5. Prove Schur’s lemma and deduce the orthonormality of irreducible characters from Theorem 1 applied to $\operatorname{Hom}(V, W)$. Then construct the character table of S_3 from scratch, using only the trivial, the sign and the permutation representations.